Statement of Purpose

Research Internship | Deep Learning & Large Language Models

I still remember the feeling of staring at a complex mathematical puzzle as a child — not frustration, but curiosity. That quiet thrill of holding multiple possibilities in my mind and slowly, logically, working toward a solution never left me. It showed up during the SOF olympiads I participated in back in school, where earning school ranks and decent zonal ranks felt less like winning and more like confirmation that this was what my mind naturally enjoyed. It showed up again in 8th or 9th grade when my school organised a basic robotics lab — nothing fancy, just enough to see that logic could move things in the physical world. Back then, I didn't know what research meant. I just knew I wanted to keep going deeper.

That instinct brought me to the Indian Institute of Engineering Science and Technology, Shibpur — one of India's recognized engineering institutions — where I am now pursuing my B.Tech in Mechanical Engineering. It might seem like a detour from machine learning, but it wasn't. Mechanical engineering taught me to think in systems: inputs, transformations, outputs, feedback loops. When I later encountered the transformer architecture for the first time, something clicked. The way attention mechanisms weigh relationships across an entire sequence — ignoring distance, focusing on relevance — felt like a mathematical poem. I had never seen logic expressed so elegantly. That moment changed what I wanted to spend my time on.

I am now deeply interested in contributing to research in deep learning, with a specific focus on large language models and computer vision. What excites me about LLMs isn't just their scale but their ability to represent reasoning in ways we don't fully understand yet. I want to work on problems that sit at that intersection — where performance meets interpretability, where models don't just predict but represent structure. My previous research internship at IIM Udaipur gave me a taste of rigorous data analysis: cleaning messy e-commerce data, identifying drivers of customer conversion through statistical and visual methods, and grounding my work in existing literature. That experience taught me that real research is slow, iterative, and often frustrating — and I enjoyed it.

My technical projects have been my real laboratory. I built an end-to-end NLP pipeline to analyze narrative patterns across hundreds of TV episodes, fine-tuning a DistilBERT model on domain-specific techniques and engineering a weighted loss function to handle class imbalance — a problem I learned to respect the hard way. I developed a RAG-based medical assistant bot using Llama-3.3-70B, Pinecone for semantic search, and custom prompt templates to prevent

hallucination, which taught me how fragile and careful LLM deployment really is. On the computer vision side, I trained a YOLOv8 model for license plate detection, squeezing 91.8% mAP out of a modest 433-image dataset through careful preprocessing and hyperparameter tuning. And with an LSTM model for Tesla stock prediction, I learned that even a 94% R^2 score doesn't mean you've understood anything — it just means you've fit the past well. That humility is probably the most important lesson any of these projects gave me.

What I want from a research internship is not a line on my resume but a real problem — something where the answer isn't known, where I have to read papers, formulate hypotheses, fail, revise, and fail better. I bring strong coding skills in Python and PyTorch, experience with the full cycle of an ML project (from data preprocessing to evaluation), and a genuine willingness to read deeply, document honestly, and persist through dead ends.

Long-term, I want to pursue graduate studies in machine learning and contribute to research that makes models more capable and more understandable. This internship is a critical step — not a formality, but a real opportunity to learn what doing research actually means before I commit to a PhD path. I am ready to work hard, ask stupid questions, and gradually ask less stupid ones.

Thank you for considering my application.